Future-safe quotients and source-sensitive obstructions

An Observational SUCC research draft

Observational SUCC research program

18 October 2026

Abstract

We give a minimal deterministic contract for discarding information while preserving declared future reports, partial actions and costs. Predictive sufficiency is separated from autonomous quotient dynamics, source-realizable paths from compatible report paths, and existential lifts from online realization. The quotient and finite path mechanisms specialize classical automata and coalgebra theory. A self-contained arithmetic construction shows that any finite ordinary-prime prefix, density one, a prime number theorem and the critical-strip zero divisor together fail to identify successor–dilation rigidity within a generalized-prime source class. Finally, we derive a boundary-aware resolvent estimate for the actual real-Gamma tail used in the signed Suzuki form. Six scoped quotient/path lemmas have been kernel checked; the arithmetic and analytic proofs are paper proofs with exact finite controls. Academic priority is unverified. No first-window Weil sign or RH proof is claimed.

1 Minimal specification and proof authority

A deterministic contract is a set X of complete source/history states, an action set A, a report map $o : X \to O$, and partial maps $t_a : X \rightharpoonup X \times \mathbb{R}_{\geq 0}$. Reports may include current source tags and the outputs of a declared proof verifier. An immutable provenance label, when required, is separately preserved by every action. Epochs and current source configurations can change inside X. A state need not admit a finite effective representation. A hypothetical path is not an actual event journal, and equality of verifier outputs does not establish soundness of that verifier.

A finite structural trace records its initial report, all reached reports, costs and the first unavailable action. Unavailability becomes an observed separator only under a specified lawful refusal/menu instrument. The statements below concern this declared structural contract; they do not authenticate a physical source or provide an undeclared instrument.

2 Exactly what can be forgotten

Write $x \simeq y$ when all finite structural traces agree. For a representation $q : X \twoheadrightarrow Y$, call q *predictively sufficient* if every finite trace is a function of $q(x)$. Call it *autonomous* if reports and partial cost-labelled actions descend to Y.

Theorem 1 (Classical quotient, with explicit contract)**.** *Predictive sufficiency is equivalent to* $\ker q \subseteq \simeq$*. Autonomy is equivalent to the following local laws whenever* $q(x) = q(y)$*:*

1. $o(x) = o(y)$*;*
2. $t_a(x)$ *is defined exactly when* $t_a(y)$ *is defined;*

3. *if* $t_a(x) = (x', c)$ *and* $t_a(y) = (y', d)$*, then* $c = d$ *and* $q(x') = q(y')$*.*

The relation $\simeq$ *is the greatest equivalence satisfying these laws. Its quotient is autonomous. Every surjective autonomous representation* q *admits a unique map* $f : Y \to X/\simeq$ *such that* $\pi = fq$*. Every executable quotient word lifts from every source representative, preserving each charged cost and the final quotient state.*

Proof. A function factors through a surjection exactly when it is constant on its fibers; apply this to each trace. If actions descend, the three local laws follow by comparing their images. Conversely define the quotient output and action using a representative; the laws make the definition independent of that choice. Comparing empty traces gives output equality. Comparing one-letter prefixes and then arbitrary suffixes gives availability, cost and successor equivalence for $\simeq$. Conversely the local laws propagate trace equality by induction on word length. Thus every such equivalence is contained in $\simeq$. Set $f(q(x)) = [x]$; this is well-defined, unique and commutes with the induced actions. Finally, a quotient transition must equal the image of the transition from any representative, so it has a lift from that representative; iterate over the word. □

This is a specialization of final semantics and quotient descent for $F(X) = O \times (1 + X \times \mathbb{R}_{\geq 0})^A$ [1, 2]; histories and source labels do not by themselves establish novelty. The repository's B15 and causal Nerode development already establish the total deterministic core. The constructive Lean theorem takes a supplied section of q and does not invoke choice. Its exact axiom audit is recorded separately.

Predictive sufficiency is strictly weaker than autonomy. Let all outputs and unit costs be identical, with $p \to u$, $q \to v$, $u \to u$, $v \to v$. Every state has the same behavior, but the representation identifying p, q and distinguishing u, v cannot update autonomously.

For bounded observations the correct state is (x, b) with residual budget b. An affordable transition of cost c leads to $(x', b-c)$; unaffordable and unavailable requests have the same rejection in this version of the contract. Applying the theorem to these augmented states gives the lawful budget quotient. Applying budget-b equivalence repeatedly to bare states would replenish resources. For example, $p \to u$, $q \to v$, $u \to u$, $v \to z$, $z \to z$, with unit costs and output one only at z, satisfies $p \simeq_1 q$ but $u \not\simeq_1 v$.

Effective existence is another boundary. In a computable unary system let $(M, n) \mapsto (M, n+1)$ and output one exactly when machine M halts within n steps. Comparison with a forever-zero loop decides nonhalting if behavioral equivalence is decidable. Thus a computable transition law does not supply a uniform algorithm for the infinite quotient. This fact has no implication about the decidability of a particular number-theoretic conjecture.

Theorem 2 (Exact finite resource threshold)**.** *Suppose* X, A *are finite, costs are nonnegative rationals, and the budget interface rejects unavailable or unaffordable requests identically at zero cost, keeps residual budget hidden, and reports the exact cost of each successful action. On unordered state pairs make a terminal edge of weight zero for differing reports; of weight* c *when only one side enables an action of cost* c*; and of weight* $\min(c, d)$ *when both enable it with unequal costs. Equal-cost actions give edges of that common cost to the successor pair. If* $D(x, y)$ *is the shortest distance to a terminal, with* $D = +\infty$ *if no terminal is reachable, then, for every rational budget* $B \geq 0$*,*

$$x \not\simeq_B y \quad \Longleftrightarrow \quad D(x, y) \leq B.$$

A finite threshold is attained and has a word witness of at most $\binom{|X|}{2}$ *actions. It can be found without expanding a residual-budget lattice. Priority for this specialized formulation is unverified.*

Proof. Before the first trace discrepancy, both runs execute the same actions with equal costs, hence follow synchronized pair edges. A discrepancy is either an output difference, a successful action versus rejection, or unequal successful cost receipts. Its minimum necessary extra budget is exactly the corresponding terminal weight. Conversely a terminal path of weight d gives a differing trace at every budget at least d: a larger budget can turn a cost-based affordability difference into unequal cost receipts, but cannot hide it. Nonnegative weights permit deleting loops. A successful path never reaches a diagonal pair, which has no possible discrepancy. Thus a simple shortest terminal path visits at most the stated number of nondiagonal pairs. The finite rational minimum is attained. □

The observation interface is load-bearing. If one compares only words lawful at both states, agreement need not be transitive: let x, y, z initially report zero, let x have a zero-reporting a-loop, let y have no a-edge, and let z have an a-edge to a one-reporting state. Then x agrees with y and y with z on their common lawful words, while x, z disagree on a. Such a relation cannot be used as a quotient equivalence without an additional testing contract.

3 Report gluing versus one lawful history

Fix a finite private state set X, a fixed original-source fiber X_σ, initial states I_σ, a report $r : X_\sigma \to Y$, and relations R_a preserving the original source. Define $y\bar{R}_a z$ by existence of an R_a edge between the corresponding private fibers. A projected word is $w = (y_0, a_0, \ldots, a_{n-1}, y_n)$ starting in $r(I_\sigma)$ and using these projected edges. Its exact private supports are

$$B_0 = I_\sigma \cap r^{-1}(y_0), \qquad B_{i+1} = R_{a_i}[B_i] \cap r^{-1}(y_{i+1}).$$

Theorem 3 (Finite private lifting and obstruction)**.** *A projected word has a private lift if and only if $B_n \neq \varnothing$, if and only if its complete private layered graph admits a nonnegative unit flow, if and only if it admits a $0/1$ path flow. Nonexistence is certified by sets C_i containing B_0 initially, closed under the specified successive transitions, and empty at the final layer. Every projected word lifts if and only if the nonempty support configurations reachable from the initial configurations form a closed invariant. If this fails, a shortest failed word has at most*

$$N_\sigma = \sum_{y \in r(X_\sigma)} (2^{|r^{-1}(y)|} - 1)$$

edges.

Proof. Induction identifies B_i with endpoints of all private lifts of the prefix. A path gives a unit flow; conservation lets one follow positive-flow edges from the supersource to the supersink. Time layers prevent cycles. Repeatedly subtracting the minimum flow on such a path gives a finite convex combination of path flows, hence integrality for this network contract. Any closed C_i contains B_i inductively, and $C_i = B_i$ provides a cut when B_n is empty. For incidence columns $e_{\text{head}} - e_{\text{tail}}$, the indicator of the source-side closed cut has $p^T A \geq 0$ and $p^T b = -1$, an exact Farkas witness. There are N_σ possible nonempty configurations (y, B). A shortest failed path cannot repeat one before its final empty update: removing the loop preserves both report compatibility and the deterministic support update. The number of nonempty prefixes, and hence of edges, is at most N_σ. Exhaustive breadth-first search proves the final alternative. □

The mechanism is classical subset construction, path decomposition and abstraction checking. The private interfaces are essential. Edges $p \to u$ and $v \to q$ with $r(u) = r(v)$ glue as reports while

$u \neq v$ prevents a lift. Extra marginal constraints can also destroy network integrality: pairwise binary anticorrelations around an odd cycle admit compatible fractional local laws but no global deterministic assignment.

Even uniform existential lifting is not an online resolver. With one report, initial state p, edges $p \xrightarrow{a,b} \{u,v\}$, $u \xrightarrow{a} \{u,v\}$ and $v \xrightarrow{b} \{u,v\}$, every finite word has a lift. Yet any resolver's first choice u or v is defeated by the next request b or a. This separates $\forall w\, \exists$lift from one causal strategy for every word, not merely from source authenticity.

4 Arbitrarily late arithmetic camouflage

Generalized primes below, in Beurling's framework [3], are positive real generators of a free commutative monoid; counting retains word multiplicities.

Theorem 4 (A uniform source nonidentification family)**.** *For every finite $H \geq 1$, there is a collision-free rational generalized-prime source agreeing with ordinary primitive, prime-power and integer data through H, with first real cell length one, $N_B(X) = X + O(\log^2 X)$, a prime count differing from the ordinary one by at most two, and*

$$\zeta_B(s) = \zeta(s)F(s), \qquad \Re s > 0,$$

as a meromorphic continuation from the Euler product on $\Re s > 1$, where F is holomorphic and nowhere zero and $F(1) = 1$. Nevertheless its rank successor and unchanged multiplication by two fail $V_2 S = S^2 V_2$.

Proof. Choose ordinary primes $p > \max(H, 2)$ and $q > p^2$. Put

$$A = p^2 - p + 1, \qquad a = \frac{p^2}{p-1}, \qquad b = \frac{qA}{q + p(p-1)},$$

and replace p, q by a, b. Word values are $ma^j b^k$ with $(m, pq) = 1$. Their p, q valuations are $2j, k$, so distinct words have distinct values. Both replacements exceed $p + 1$; consequently the source through $p + 1$ is $1, \ldots, p - 1, p + 1$. For $x = (p - 1)/2$ the difference

$$\operatorname{rank}(V_2 S x) - \operatorname{rank}(V_2 x) - 2 = -1$$

exhibits the failed braid, while every bounded operation staying through H is unchanged.

Let $C = (1 - 1/p)(1 - 1/q)$ and $R(y) = \lfloor y \rfloor - \lfloor y/p \rfloor - \lfloor y/q \rfloor + \lfloor y/(pq) \rfloor$. Then $|R(y) - Cy| \leq 2$ and direct algebra gives $(1 - 1/a)(1 - 1/b) = C$. Thus

$$N_B(X) = \sum_{a^j b^k \leq X} R(X a^{-j} b^{-k}).$$

Writing $J = \max\{j : a^j \leq X\}$ and $K = \max\{k : b^k \leq X\}$, sum the rounding error over at most $(J + 1)(K + 1)$ terms. The omitted geometric tail, split at $j = J$, is at most $(J + 1)(1 - 1/a) + 1$ after multiplication by CX. Hence

$$|N_B(X) - X| \leq 2(J + 1)(K + 1) + (J + 1)(1 - 1/a) + 1.$$

Only two primitive values are replaced, proving the prime-count bound. On $\Re s > 1$ the absolutely convergent Euler products give

$$F(s) = \frac{(1 - p^{-s})(1 - q^{-s})}{(1 - a^{-s})(1 - b^{-s})}.$$

Every factor is holomorphic and nonzero on $\Re s > 0$; ordinary zeta continuation and $F(1) = 1$ prove the rest. In particular the critical-strip zero divisor and pole residue agree without using any zero ordinate. □

A terminating adaptive protocol using only bounded source-prefix queries has the same transcript on a sufficiently late member of this family as on ordinary arithmetic. Even the displayed global summaries do not break that ambiguity. This does not exclude a finite proof of an authenticated universal source law. Nor does the construction preserve the standard completed functional equation, canonical Gamma term or full Weil distribution. The generalized-prime framework is classical; scholarly priority for this specific simultaneous construction remains unverified.

5 Retaining the authentic Gamma tail

On $I = (-1/2, 1/2)$ set $d_j = 4j + 1$, $\alpha_j = d_j/2$ and

$$(E_\alpha h)(x) = \int_I e^{-\alpha|x-y|} h(y)\mathrm{d}y, \qquad \Gamma_j h = \frac{4}{d_j} h - E_{\alpha_j} h.$$

These are the rungs of the one real-Gamma source in Suzuki's completed form [4]. They do not replace its signed prime, pole or contact terms. Let $S_r(J) = \sum_{j \geq J} d_j^{-r}$ and $a_J = 16 S_3(J)$.

Theorem 5 (Endpoint-paid tail expansion)**.** *If $J \geq 1$ and $h \in H^4(I) \cap H_0^2(I)$, then*

$$\sum_{j \geq J} \Gamma_j h = -a_J h'' + e_J,$$

where, writing $d = 4J + 1$ and $L_J(u, v) = (|u|^2 + |v|^2 + 2|u\bar{v}|de^{-d/2})^{1/2}$,

$$\begin{aligned} \|e_J\|_2 \leq\ & 8 L_J(h''(-1/2), h''(1/2)) \sqrt{S_3 S_4} \\ & + 16 L_J(-h'''(-1/2), h'''(1/2)) \sqrt{S_4 S_5} + 64 S_5 \|h''''\|_2. \end{aligned}$$

For real endpoint values the cross term may be sharpened from $|u\bar{v}|$ to $\max(uv, 0)$. The quadratic form also satisfies

$$\langle \Gamma_{\geq J} h, h \rangle \geq 16 S_3 \|h'\|_2^2 - 64 S_5 \|h''\|_2^2.$$

Proof. Since $h = h' = 0$ at the endpoints, two integrations by parts give $\Gamma_j h = -\alpha_j^{-2} E_{\alpha_j} h''$. Two more yield

$$\begin{aligned} \Gamma_j h = \ & -16 d_j^{-3} h'' - 16 d_j^{-4} E_{\alpha_j} h'''' \\ & + e^{-\alpha_j (x+1/2)} (8 d_j^{-3} h''(-1/2) - 16 d_j^{-4} h'''(-1/2)) \\ & + e^{-\alpha_j (1/2-x)} (8 d_j^{-3} h''(1/2) + 16 d_j^{-4} h'''(1/2)). \end{aligned}$$

The exponential operator has norm at most $4/d_j$. The squared norm of an endpoint pair is $(|u|^2 + |v|^2)(1 - e^{-d_j})/d_j + 2\Re(u\bar{v}) e^{-d_j/2}$. Since $d_j e^{-d_j/2}$ decreases for $d_j \geq 5$, the stated common pair bound follows. Sum the rung norms and use $\sum d_j^{-7/2} \leq \sqrt{S_3 S_4}$ and $\sum d_j^{-9/2} \leq \sqrt{S_4 S_5}$. For the quadratic bound, zero extension belongs to $H^2(\mathbb{R})$. The Fourier multiplier of a rung is $16\xi^2/[d_j(d_j^2 + 4\xi^2)]$; apply $(1 + t)^{-1} \geq 1 - t$ and Plancherel. □

The endpoint contribution is $O(J^{-5/2})$ and the bulk remainder $O(J^{-4})$, for fixed h. The estimate does not impose the false extra conditions $h'' = h''' = 0$ at the endpoints. Exact rational series enclosures and outward square roots implement it. It reduced tail-action uncertainty on all four tested $J = 24$ columns, but the retained action also moves the residual center: a smaller error bound alone need not improve a signed Schur lower bound.

Let F_c be the original four-dimensional low vector with coefficients c, E the original primitive high subspace, Q_E its orthogonal projection, and H_∞ the original self-adjoint signed source operator. For a trial $h_c = F_c - x_c$ with $x_c \in E$ in the valid common form domains and h_c in the operator domain, and inherited original high coercivity $D \geq \kappa I$ in the correct primitive Hilbert space, the valid bound is

$$S_4[c] \geq q(h_c) - \kappa^{-1} \|Q_E H_\infty h_c\|_2^2.$$

The new estimate controls part of this expression; it proves neither its sign nor every-radius positivity. Paired source trials remain ambiguous. The complete signed source, one real Gamma, both polar signs, contact and genuine prime half-density remain fixed throughout.

6 Reproducibility and status

This paper accompanies the branch `research/academic-transmutation-2026-10-18`, based on draft PR177 commit `f309d9104ba480c1999c962af1df7c50f19e1bfe`. The original signed source blob is `0cc8510b71d178fc79e8d410a86d8b6014cf9178`. The accompanying dossiers supply complete domains, exact scripts, independent finite checkers, mutation controls, primary-source comparisons and run receipts. Lean 4.14.0 checks the quotient/path fragment, not the analytic or arithmetic theorems, not the Python programs and not the whole framework. The finite certificates establish statements about their supplied tables. The source nonidentification and Gamma estimates are self-contained paper derivations. All academic priority claims remain unverified. RH remains open.

References

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